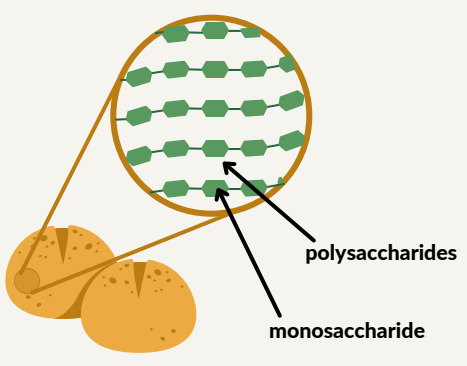
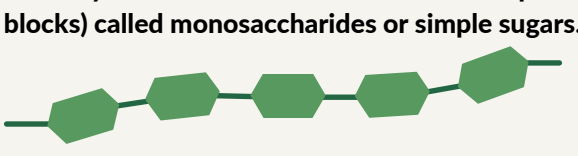
CARBOHYDRATES

A Closer Look at your carbs

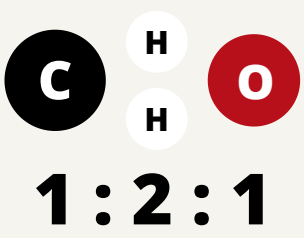


Carbohydrates are macromolecules made up of monomers (the building blocks) called monosaccharides or simple sugars.



Examples of monosaccharides: glucose, fructose, galactose

All monosaccharides contain exclusively carbon, hydrogen, and oxygen atoms with a ratio of 1:2:1 respectively.



Disaccharides are made up of 2 monosaccharides.
Examples of disaccharides: sucrose, maltose, lactose

Polysaccharides are made up of multiple monosaccharides. They are the most common form of carbohydrates.
Examples of polysaccharides: starch, glycogen, cellulose

A CARB'S JOURNEY

Carbohydrates are found in almost everything we eat, and that is how we get those much needed carbohydrates into our body[1]. They come all sorts of shapes and sizes, including, but not limited to:

SUGARS

FIBRES

STARCHES

They provide us with the energy we need to perform essential bodily functions, as well as other physical activity.

Healthy Carbs

- unprocessed or minimally processed foods

whole grains

vegetables

fruits

beans

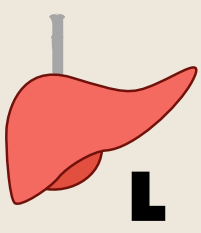
Unhealthy Carbs

- highly processed or refined foods
- easily digested and turned into energy faster

white bread

pastries

soft drinks



THE LIVER

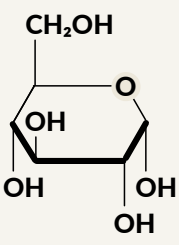
The liver is capable of storing glucose when necessary[2]. When the body ingests these carbohydrates, the liver stores them in the form of glycogen, another carbohydrate. However, when energy is needed, the liver will turn the glycogen polysaccharide into glucose monosaccharides through a process called glycogenolysis, by breaking up all the glycosidic bonds.

Sugar can also be made by the body through harvesting amino acids, waste products, and fat byproducts through a process called gluconeogenesis*.

Glucose

a.k.a. "blood sugar"

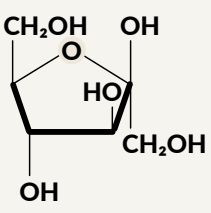
Energy is **stored in the liver** as glucose[2]. Even if a different simple sugar enters the body, **it will be turned into glucose first** before becoming useful to cells to generate energy.



Fructose

a.k.a. "nature's candy"

Fructose is found in **berries and fruits**, making up 5-8% of their total weight, as well as 40% of honey[3]. Unlike glucose, it is **metabolized independently of insulin**, so fructose syrups and sweeteners are safe for diabetics[4].



Fructose is absorbed in the intestines just like other nutrients. The liver **converts fructose into pyruvates much quicker** than other simple sugars, so blood fructose levels are generally much lower[5].

However, the absorption of fructose is still not very well understood. The **absorptive capacity** for fructose when not found in sucrose is **relatively small**, with a low Glycaemic index** rating of 19, and can cause negative symptoms for those with fructose intolerance.

Symptoms of incomplete fructose absorption like bloating result from the osmotically-induced distension of the small intestine, even without the fermentation of carbohydrates producing excess hydrogen gas.

The ingestion of a large amount of fructose can change the body's metabolic abilities, as it will bypass the regulatory mechanism in glycolysis. It can replace glucose as a source of fuel, which may also temporarily cause poor glucose absorption. It will also shift the fatty acid metabolism process from oxidation to the esterification of free fatty acids.

Around the World in Half an Hour[6]

A CARB'S JOURNEY PT. 2

MOUTH

The teeth mash up the food and the saliva begins the process of breaking it down.

"Sweet Lips, Tennessee, United States of America"

STOMACH & INTESTINES

The chewed up food is mixed with enzymes in the stomach, which breaks down the polysaccharides into monosaccharides.

The majority of the monomers are then absorbed through the intestinal mucosa of the small intestines. From there, the nutrients may go either to the liver for storage purposes, or directly into the bloodstream.

"Darmstadt, Germany" (translation: intestine town/city)

LIVER

Glucose can be stored here as glycogen, then produced and delivered to cells for energy.

"Liverpool, England"

BLOODSTREAM

Glucose is delivered to the body's cells.

"Stand Off, Kainai 'Blood' Nation, Alberta, Canada"

CELL

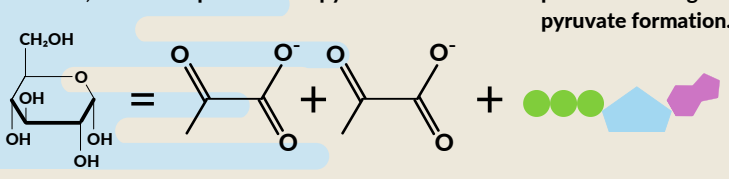
The glucose enters the cytosol and begin oxidizing, which starts the process of cellular respiration. The majority of the process happens and ends in the mitochondrion, thus producing energy in the overall reaction.

"Xibao (西堡), China" ("cell" also has the pinyin "xibao" in Mandarin)

Cellular Respiration[7]

A CARB'S JOURNEY PT. 3

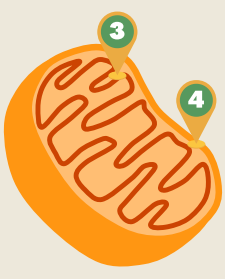
1 The glucose molecules go through glycolysis, in which they are split into two pyruvate molecules. Other sugars that are not glucose are first converted to a sugar intermediate, then also split into two pyruvate molecules.



2 Two activated carrier molecules***, ATP and NADH, are produced during the pyruvate formation.

3 The pyruvate molecules pass into the mitochondria, where they are converted into CO_2 and a two-carbon acetyl group that will be attached to another enzyme to create more activated carrier molecules.

4 The remaining steps in the oxidative breakdown of the carbohydrate and fat molecules in the mitochondria is what releases most of the energy that will be harnessed to produce ATP for the cell.



DIABETIC NEPHROPATHY

Diabetes occurs when the pancreas does not produce insulin, a hormone that plays an integral part in processing glucose, properly. In the case of insulin resistance, the liver doesn't recognize the insulin in the body and continues to make ridiculous amounts of glucose. If not enough insulin is produced by the body, it can result in the release of free fatty acids, resulting in ketoacidosis** and other extremely dangerous conditions affecting the whole body.**

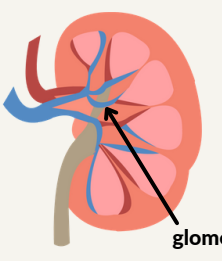
DIABETIC NEPHROPATHY

is a kidney-related complication of diabetes caused by prolonged hyperglycemia (unregulated high blood sugar levels, resulting from excessive glucose in the bloodstream)[8].

The kidney's delicate filtering system is slowly damaged over many years. If left untreated, it may progress to kidney failure (end-stage kidney disease). At this stage, treatment options are dialysis or a kidney transplant.

However, kidney disease can be prevented or slowed through early treatment and leading a healthy lifestyle, which reduces the chance of additional complications.

25% of diabetics eventually develop kidney disease.



The kidney is the body's filtering system. It contains millions of tiny blood vessel clusters called glomeruli that filter waste from the blood. Severe damage to the glomeruli decreases the kidney's ability to function[9].

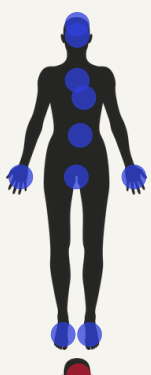
Diabetic nephropathy happens when the blood pressure rises due to the diabetes. This in turn increases the pressure on the delicate glomeruli and damages them.

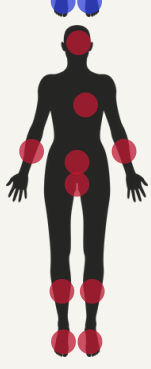
Signs and symptoms may not show in early stages, but will become more prominent as time progresses:

- worsening blood pressure control
- protein in urine, increased need to urinate
- swelling of extremities and eyes
- reduced need for insulin or diabetes medication
- confusion or difficulty concentrating
- shortness of breath
- loss of appetite, nausea, and vomiting
- persistent itching
- fatigue

More serious, and possibly irreversible complications may also develop over months or years:

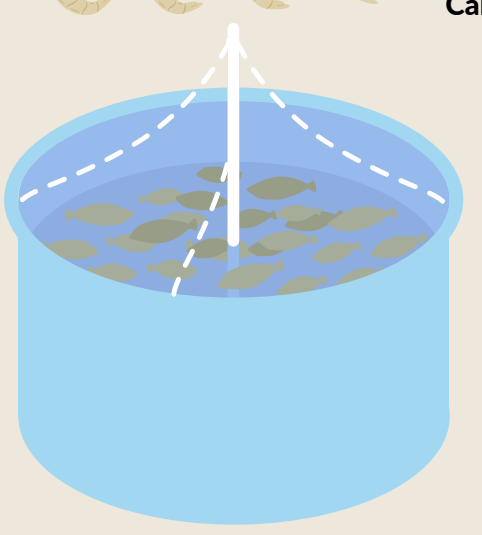
- fluid retention (may cause swelling of appendages, high blood pressure, pulmonary edema)
- hyperkalemia, anemia
- cardiovascular disease, stroke
- diabetic retinopathy
- pregnancy complications
- damaged nerves and blood vessels (may cause foot sores, erectile dysfunction, diarrhea)





BIOFLOC

Carbs in Sustainable Water Treatment Systems



The **addition of carbohydrates** to an aquaculture production system provides an **energy source** to the microbial organisms that participate in the system, allowing them to **convert ammonia and nitrate into microbial mass**. Not only does this process help **lower ammonia and nitrite levels**, reducing the need for water exchange; the microbial biomass generated also serves as food for the culture species, **increasing feed use efficiency**.

BIOFLOC SYSTEMS

are aquaculture production systems that use organisms suspended in a matrix with particulate organic matter that can be used for the improvement of water quality and performing waste treatment and disease prevention. They may use bacteria, algae, fish, shrimp, protozoa, or other marine species to participate in the system.

Effects of carbohydrate sources on a biofloc shrimp nursery[10]

The study showed that the addition treatments of both cornstarch and molasses resulted in low ammonium and nitrogen levels in the water.

Not only did the addition of carbohydrates enhance the growth of the participating organisms, they also improved the immunological responses of the participating species, Pacific white shrimp (*L. vannamei*).

Through the experiment, it was found that the corn starch addition treatment yielded significantly better shrimp production than the molasses.

Monday March 29, 2021

Pacific white shrimp (*Litopenaeus vannamei*)

[1]"Carbohydrates." The Nutrition Source, Harvard T.H. Chan School of Public Health, 22 May 2019, www.hsph.harvard.edu/nutritionsource/carbohydrates/.
[2]Pointer, Kathleen. "What Is Glucose and What Does It Do? - Healthline." Reviewed by Elaine K. Luo, Healthline, Healthline, 24 Mar. 2017, www.healthline.com/health/glucose.
[3]Ziesenitz, S. C. "Authorised EU Health Claim for Fructose." Foods, Nutrients and Food Ingredients with Authorised EU Health Claims, vol. 2, 2015. ScienceDirect, Research Database, doi:10.1016/b978-1-78242-382-9.00011-6.
[4]Raman, M. "Testing for Chronic Diarrhea." Advances in Clinical Chemistry, 2017. ScienceDirect, Research database, doi:10.1016/bs.acc.2016.09.004.
[5]Von Rymon Lipinski, G.-W. "Reduced-Calorie Sweeteners and Caloric Alternatives." Optimising Sweet Taste in Foods, 2006. ScienceDirect, Research database, doi:10.1533/9781845691646.2.252.
[6]"How Our Bodies Turn Food into Energy." Reviewed by Meredith Cotton, How Our Bodies Turn Food Into Energy | Kaiser Permanente Washington, 3 Jan. 2019, wa.kaiserpermanente.org/healthAndWellness?item=%2Fcommon%2FhealthAndWellness%2Fconditions%2Fdiabetes%2FfoodProcess.html.
[7]Alberts, B., et al. "How Cells Obtain Energy from Food." Molecular Biology of the Cell, 4th ed., Garland Science, 2002. National Center for Biotechnology Information, www.ncbi.nlm.nih.gov/books/NBK26882/.
[8]"Diabetic Nephropathy." Mayo Clinic, Mayo Foundation for Medical Education and Research, 19 Sept. 2019, www.mayoclinic.org/diseases-conditions/diabetic-nephropathy/symptoms-causes/syc-20354556.
[9]"The Liver & Blood Sugar." Diabetes Education Online, Diabetes Teaching Center at the University of California, San Francisco, dtc.ucsf.edu/types-of-diabetes/type1/understanding-type-1-diabetes/how-the-body-processes-sugar/the-liver-blood-sugar/.
[10]Tinh, Tran Huu, et al. "Effects of Carbohydrate Sources on a Biofloc Shrimp Nursery." Global Aquaculture Advocate, Global Aquaculture Alliance, 29 Mar. 2021, www.aquaculturealliance.org/advocate/effects-of-carbohydrate-sources-on-a-biofloc-shrimp-nursery/.

*gluconeogenesis: the metabolic process by which cells produce sugars from non-carbohydrate precursors for catabolic reactions. Glucose is the primary sugar product since it is the only energy source used by many body parts (most notably the brain), with the exception of ketone bodies during fasting periods
**Glycaemic index: a rating system for foods with carbohydrates that shows how quickly each food affects the blood sugar level when the food is eaten on its own. A number from 1 to 100 is assigned with glucose at the highest at 100, representing the relative rise in the blood glucose level after a certain period of time.
***activated carrier molecules: molecules that can be split to release free energy, but only if there is an excess of the molecule relative to its equilibrium concentration. Key examples are: ATP, GTP, NADH, FADH2, and NADPH.
****ketoacidosis: a life threatening condition in which the body starts breaking down fat too quickly into ketones, causing the blood pH to drop. It most commonly affects diabetics.